



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°7_ – SYSTEME

COURS: SYSTEME

PRÉNOM: Lens Sandro

NOM: PÉTIOTE

NIVEAU: L3

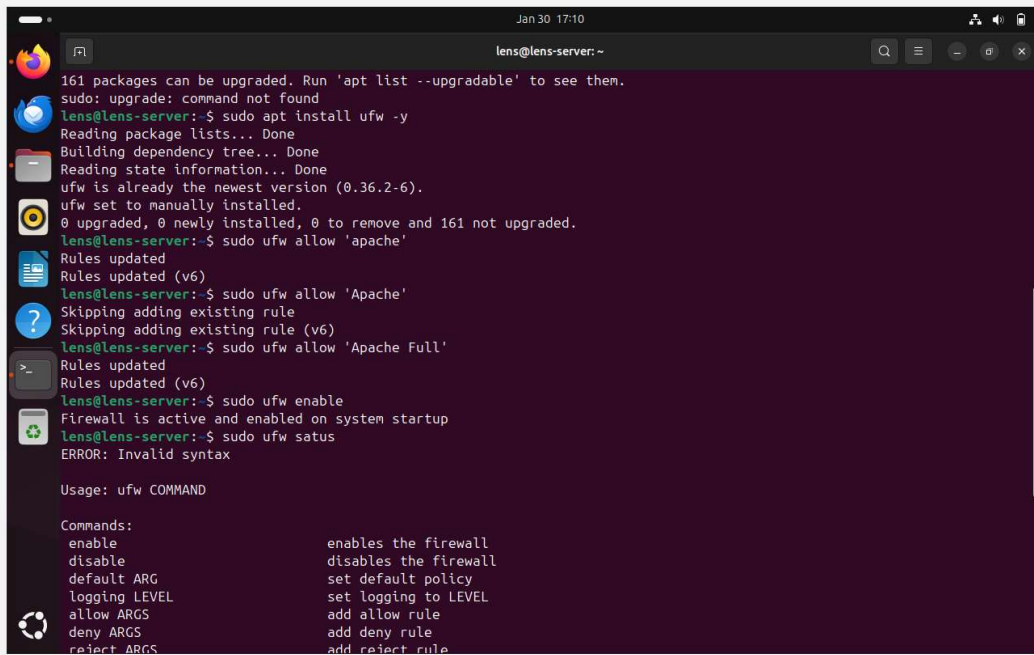
DATE: 29/01/2026

L'objectif de ce TD est de:

1. Installer et configurer un serveur Web Apache sous Ubuntu
2. Déployer un site web statique (HTML/CSS/JS)
3. Créer et activer un VirtualHost Apache

1. Déployer un Serveur Web et Héberger un Site Statique

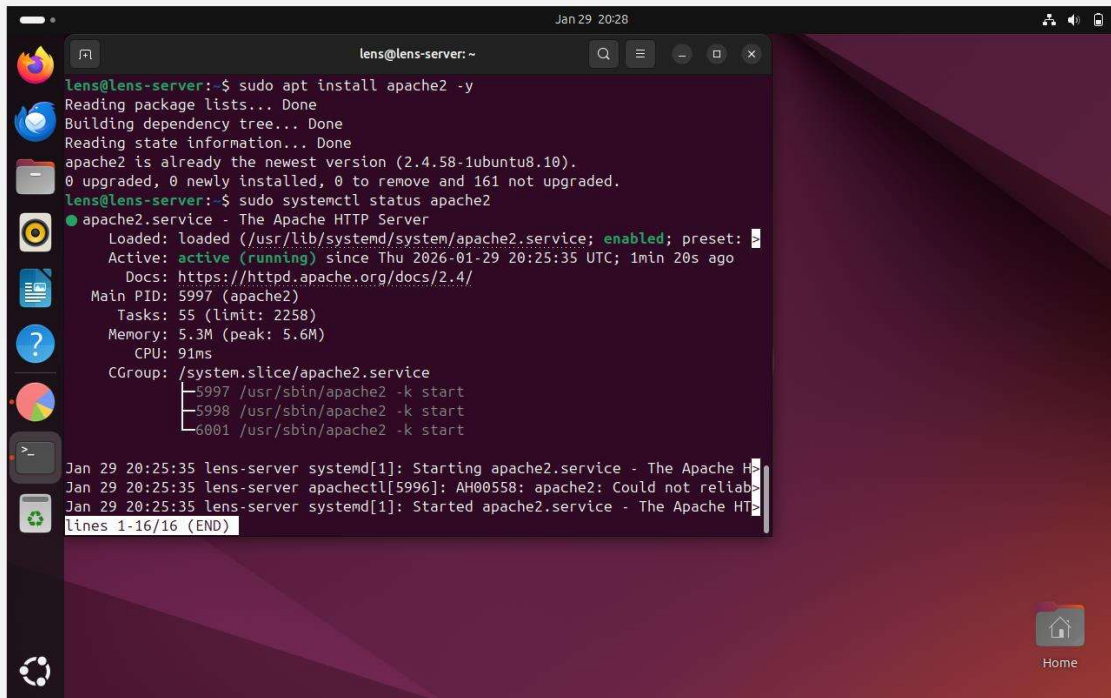
Configuration du pare-feu et Installation du serveur Apache et verification du service



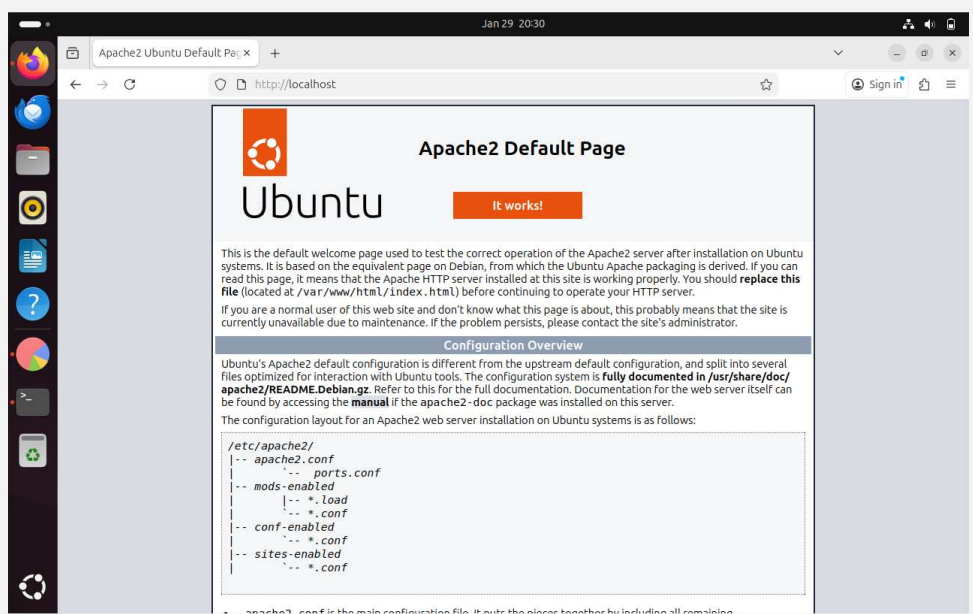
```
Jan 30 17:10
lens@lens-server: ~
161 packages can be upgraded. Run 'apt list --upgradable' to see them.
sudo: upgrade: command not found
lens@lens-server:~$ sudo apt install ufw -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.2-6).
ufw set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 161 not upgraded.
lens@lens-server:~$ sudo ufw allow 'apache'
Rules updated
Rules updated (v6)
lens@lens-server:~$ sudo ufw allow 'Apache'
Skipping adding existing rule
Skipping adding existing rule (v6)
lens@lens-server:~$ sudo ufw allow 'Apache Full'
Rules updated
Rules updated (v6)
lens@lens-server:~$ sudo ufw enable
Firewall is active and enabled on system startup
lens@lens-server:~$ sudo ufw satus
ERROR: Invalid syntax

Usage: ufw COMMAND

Commands:
enable          enables the firewall
disable         disables the firewall
default ARG     set default policy
logging LEVEL   set logging to LEVEL
allow ARGS      add allow rule
deny ARGS       add deny rule
reject ARGS     add reject rule
```

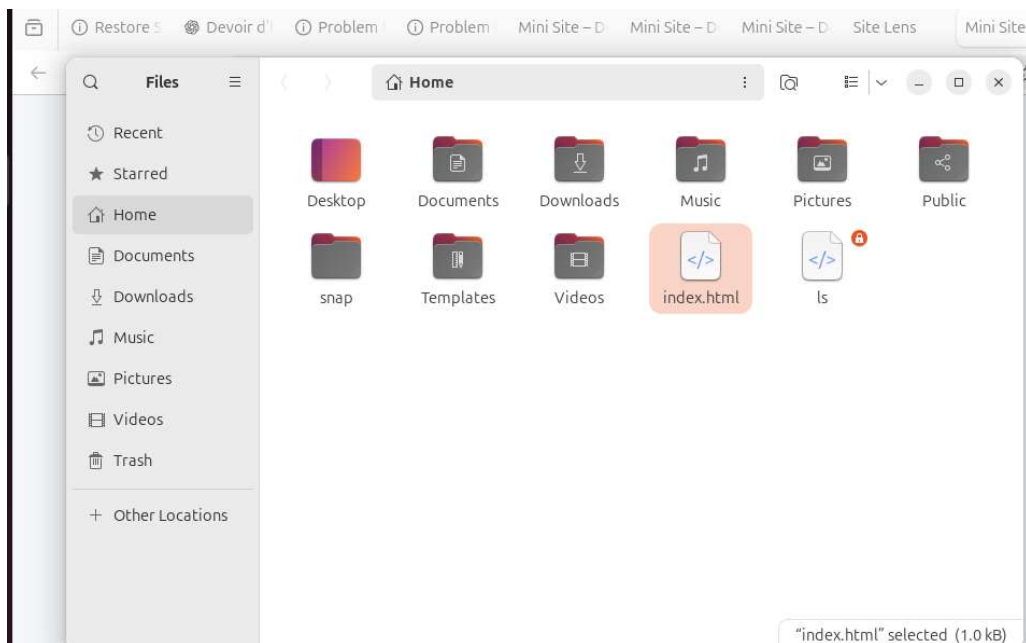


```
lens@lens-server: ~  
lens@lens-server:~$ sudo apt install apache2 -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
apache2 is already the newest version (2.4.58-1ubuntu8.10).  
0 upgraded, 0 newly installed, 0 to remove and 161 not upgraded.  
lens@lens-server:~$ sudo systemctl status apache2  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset:   
   Active: active (running) since Thu 2026-01-29 20:25:35 UTC; 1min 20s ago  
   Docs: https://httpd.apache.org/docs/2.4/  
   Main PID: 5997 (apache2)  
     Tasks: 55 (limit: 2258)  
    Memory: 5.3M (peak: 5.6M)  
       CPU: 91ms  
   CGroup: /system.slice/apache2.service  
           └─5997 /usr/sbin/apache2 -k start  
             └─5998 /usr/sbin/apache2 -k start  
               └─6001 /usr/sbin/apache2 -k start  
  
Jan 29 20:25:35 lens-server systemd[1]: Starting apache2.service - The Apache H  
Jan 29 20:25:35 lens-server apachectl[5996]: AH00558: apache2: Could not reliab  
Jan 29 20:25:35 lens-server systemd[1]: Started apache2.service - The Apache HT  
lines 1-16/16 (END)
```



Création d'un répertoire pour mon site.

```
lens@lens-server:~$ sudo cp index.html /var/www/html/
[sudo] password for lens:
cp: cannot stat 'index.html': No such file or directory
lens@lens-server:~$ pwd
/home/lens
lens@lens-server:~$ ls
Desktop Documents Downloads ls Music Pictures Public snap Templates Videos
lens@lens-server:~$ nano index.html
lens@lens-server:~$ sudo cp index.html /var/www/html
lens@lens-server:~$ sudo nano /etc/apache2/sites-available/monsite.conf
lens@lens-server:~$ sudo a2ensite monsite.conf
Enabling site monsite.
To activate the new configuration, you need to run:
systemctl reload apache2
lens@lens-server:~$ sudo systemctl reload apache2
```



Affichage des codes de mon site dans le fichier index.html

```
lens@lens-server:~$ cat /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
  <title>Site Lens</title>
</head>
<body>
  <h1>Bienvenue sur mon site</>
  <p>Ce site est heberge avec apache</>
</body>
</html>
lens@lens-server:~$
```

Configuration Apache (VirtualHost)

```
lens@lens-server: ~
GNU nano 7.2 /etc/apache2/sites-available/monsite.conf
<VirtualHost *:80>
    serverName monsite.local
    DocumentRoot /var/www/monsite
    <Directory /var/www/monsite>
        AllowOverride All
    </Directory>
</VirtualHost>

# The ServerName directive sets the request scheme, hostname and port that
# the server uses to identify itself. This is used when creating
# redirection URLs. In the context of virtual hosts, the ServerName
# specifies what hostname must appear in the request's Host: header to
# match this virtual host. For the default virtual host (this file) this
# value is not decisive as it is used as a last resort host regardless.
# However, you must set it for any further virtual host explicitly.
#ServerName www.example.com

ServerAdmin webmaster@localhost
DocumentRoot /var/www/html

# Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
# error, crit, alert, emerg.
# It is also possible to configure the loglevel for particular
# modules, e.g.
#LogLevel info ssl:warn

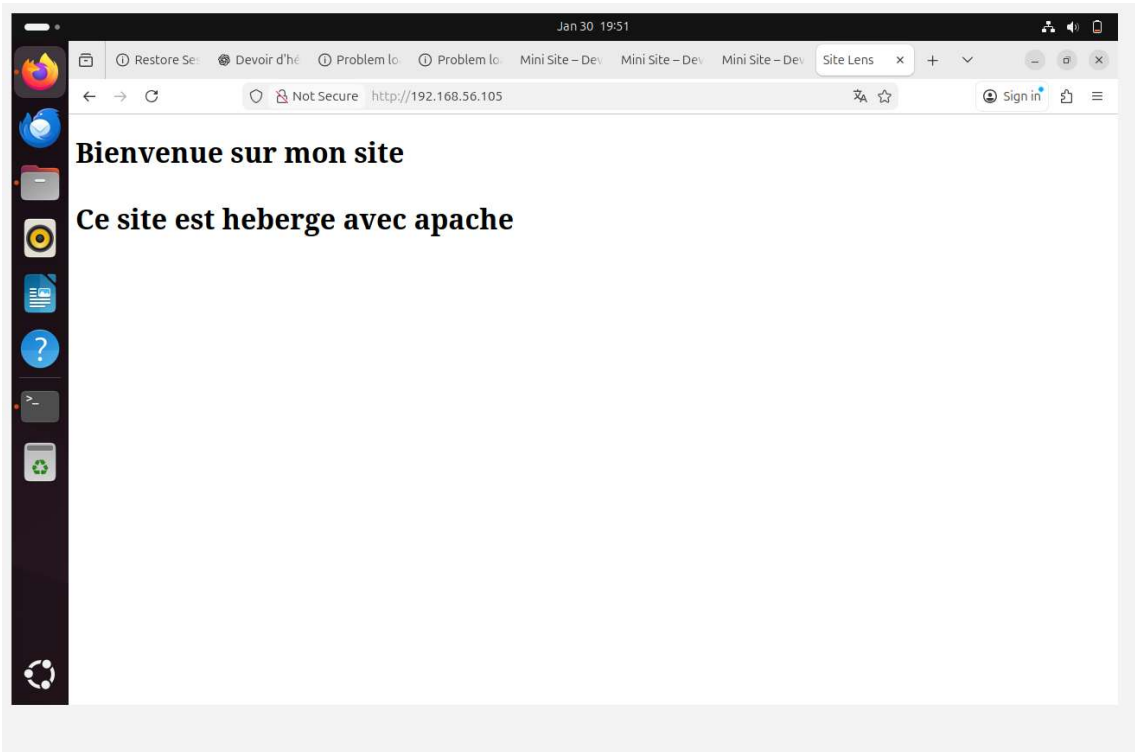
ErrorLog ${APACHE_LOG_DIR}/error.log
CustomLog ${APACHE_LOG_DIR}/access.log combined

[ Read 35 lines ]
⌵ Help ⌵ Write Out ⌵ Where Is ⌵ Cut ⌵ Execute ⌵ Location ⌵ Undo
```

Activation du site et recherche de l'address ip.


```
lens@lens-server: ~  
bash: /etc/apache2/sites-available/: Is a directory  
lens@lens-server:~$ sudo a2ensite monsite.conf  
Site monsite already enabled  
lens@lens-server:~$ sudo systemctl reload apache2  
lens@lens-server:~$ sudo chown -R www-dta:www-data /var/www/html/  
chown: invalid user: 'www-dta:www-data'  
lens@lens-server:~$ sudo chmod -R 755 /var/www/html/  
lens@lens-server:~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:6b:6c:c5 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3  
        valid_lft 80544sec preferred_lft 80544sec  
    inet6 fd17:625c:f037:2:8dc4:ee10:22f3:1199/64 scope global temporary dynamic  
        valid_lft 86071sec preferred_lft 14071sec  
    inet6 fd17:625c:f037:2:a00:27ff:fe6b:6cc5/64 scope global dynamic mngtmpaddr  
        valid_lft 86071sec preferred_lft 14071sec  
    inet6 fe80::a00:27ff:fe6b:6cc5/64 scope link  
        valid_lft forever preferred_lft forever  
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:1d:e3:15 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.56.105/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s8  
        valid_lft 434sec preferred_lft 434sec  
    inet6 fe80::5b14:5d4:dd4a:f49f/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
lens@lens-server:~$ ls /var/www/html/  
index.html  
lens@lens-server:~$ cat /var/www/html/index.html
```

Test avec l'address ip dans firefox



Cocclusion.

Ce travail m'a permis de comprendre comment installer et configurer un serveur Web Apache sous Ubuntu, savoir déployer un site web statique (HTML/CSS/JS) et comment créer et activer un VirtualHost Apache. Donc la tâche est réussie.